|INCH-POUND|

MIL-L-23251E
29 September 1989
SUPERSEDING
MIL-L-23251D
12 November 1988

MILITARY SPECIFICATION
LATHES, ENGINE AND TOOLROOM
2516 AND 3220

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers electric motor driven, engine lathes and toolroom lathes for use in general manufacturing or toolroom type operations with equipment and accessories specified herein. The lathes shall have geared head drive with quick change feeds and either manual shift or variable speed headstock. The requirements set forth in this specification and the completed Ordering Data (see 3.5 and 6.2) establish the Government's minimum requirements for each procurement.

1.2 Classification. The lathes covered by this specification shall be of the following types, sizes, and classes. The type, size, and class to be furnished shall be as specified (see 6.2).

Type I - Engine lathes
  Size 2516 - 25-inch swing over bed - 16-inch swing over cross slide
    Class 1 - 20 to 30 horsepower
    Class 2 - 15 to 20 horsepower
  Size 3220 - 32- inch swing over bed - 20-inch swing over cross slide
    Class 1 - 30 to 40 horsepower
    Class 2 - 40 to 50 horsepower
Type 11 - Toolroom lathes
  Size 2516 - 25-inch swing over bed - 16-inch swing over cross slide
    20 to 30 horsepower
  Size 3220 - 33-inch swing over bed - 27-inch swing over cross slide
    Class 1 - 30 to 40 horsepower
    Class 2 - 40 to 50 horsepower

|Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, 2163 Airways Boulevard, Memphis, Tennessee 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.|

AMSC N/A FSC 3416

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:02Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interferences.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Unless otherwise indicated, copies of Federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

U. S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402-0001.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:02Z
Check the source to verify that this is the current version before use.

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN GEAR MANUFACTURERS' ASSOCIATION (AGMA)

AGMA 390.03a - Gear Classification, Materials and Measuring Methods for Bevel, Hypoid, Fine Pitch Wormgearing and Racks Only as Unassembled Gears. (DOD adopted)

AGMA 2000-A88 - Gear Classification and Inspection Handbook. (DOD adopted)

AGMA 2001-B88 - Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth.

(Application for copies should be addressed to the American Gear Manufacturers' Association, Standards Department, 1500 King Street, Suite 201, Alexandria, VA 22314-2717.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B5.8 - Chucks and Chuck Jaws. (DOD adopted)

ANSI B5.9 - Spindle Noses for Toolroom Lathes, Engine Lathes, Turret Lathes, and Automatic Lathes. (DOD adopted)

ANSI B5.10 - Machine Tapers. (DOD adopted)

ANSI 85.16 - Accuracy of Engine and Toolroom Lathes. (DOD adopted)

ANSI B11.6 - Safety Requirements for the Construction, Care, and Use of Lathes. (DOD adopted)

ANSI/ISO 54 - Cylindrical Gears For General Engineering and for Heavy Engineering, Modules and Diametral Pitches of.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018-3363.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEW MG 1 - Motors and Generators. (DOD adopted)

Source: https://assist.dla.mil -- Downloaded: 202610-19T16:02Z
Check the source to verify that this is the current version before use.

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, N.W., Washington, DC 20037-1580.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery. (DOD adopted)

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of Precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), one complete lathe shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

3.2 Design. The machine shall be new (not a prototype) and one of the manufacturer's current production models capable of operations in accordance with the requirements herein. The lathe shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be readily accessible for adjustment, replacement, or repair. Neither a gap bed design, nor the use of a spacer between the headstock and the bed shall be acceptable in order to achieve the capacity required in this specification.

3.2.1 Measuring and indicating device calibrations. Measuring and indicating devices shall be graduated in inch-pound (see 6.5.1), metric (see 6.5.2), or dual (inch-pound and metric) units as specified by the procuring activity (see 6.2). Regardless of the measurement system used, all measuring and indicating devices on the lathe shall be graduated in the same system.

3.2.1.1 Dials. Rotating dials or handwheels shall be used for longitudinal movement and shall be graduated in increments not larger than 0.005 inch. Rotating dials or handwheels shall be used for cross motion and shall be graduated in increments not greater than 0.001 inch. Such graduations shall accurately indicate machine component movement or material removal. Dial and handwheel circumferences shall be permanently and legibly engraved or etched on a nonglare background with graduations that can be read from the operator's normal position. All feed dials shall have independent zero adjustments and shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:02Z
Check the source to verify that this is the current version before use.

3.2.1.2 Digital readout system. Unless otherwise specified (see 6.2), a digital readout system shall be installed on the X and Z axes of travel. The system shall consist of a position feedback device, an electronic display for each axis, and all related components, mounting hardware, and electrical equipment necessary to ensure complete operation of the system. The system shall provide a readily visible digital display over the full working range of each axis on which installed. The X axis and the Z axis resolution shall be 0.0002 inch. The system accuracy shall be 0.0010 inch verified upon installation and calibration of each system on the machine. The repeatability shall be 0.0001 inch, measured over the full travel on each axis. The X axis shall show twice the amount of cross slide movement to reflect the reduction of the workpiece diameter. Special requirements for the system components shall be as specified (see 6.2). The display shall indicate axis location in both the inch-pound and metric units. An input power source fluctuation of ±10 percent shall not adversely affect the digital readout system operation.

3.2.1.2.1 Position feedback device. A position feedback device including all related components, hardware, and electrical equipment shall be provided for each axis having a readout system. The feedback device shall be of the independent type with a resolution equal to the resolution of the digital readout system. Each feedback device shall be installed with a cover or guard to prevent physical damage to any portion of the feedback device. The connecting cable from the position feedback device shall be electronically shielded to protect against electronic interference and shall be armored to protect against contamination and damage. The device shall provide axis travel feedback over the entire range of axis travel for each specified axis. Installation of the position feedback device shall not limit machine travel on any axis nor eliminate manual machine stops. Provisions shall be made to prevent damage to the feedback device in the event of overtravel. The feedback unit and related components shall be protected from contamination. Position feedback devices shall be easily and quickly interchangeable between like machines procured under the same contract.

3.2.1.2.2 Electronic display console. The electronic display console for the digital readout system shall contain a universal digital display with individual illuminated digits of not less than 0.43 inch in height. Digits shall be high efficiency displays for high light ambient conditions with wide viewing angle and shall be readable at a distance of not less than 16 feet. The console shall display at least six digits per axis simultaneously. The display console shall also provide keypad entry, memory, absolute/incremental counting, data preset, power failure indicator, inch/metric conversion, and absolute zero recal 1. The electronic display console shall be installed so as to be readily accessible and easily read from the operator's normal work station. The console shall be mounted to withstand shock and vibration generated by the machine. Each connecting cable shall be armored and shall be securely fastened to the structure of the machine. Location of installed cables shall prevent any interference with normal machine operation. Electronic display consoles shall be easily and quickly interchangeable between like machines procured under the same contract.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials to the maximum extent possible provided such materials will not jeopardize the

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:02Z
Check the source to verify that this is the current version before use.

intended use, performance, or design life of the machine. Reclaimed materials shall have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used, rebuilt, or remanufactured end products are allowed under this specification.

3.2.3 Energy efficiency. The machine and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency in accordance with the developments available within the industry.

3.2.4 Controls. All mechanical, hydraulic, and pneumatic operating controls shall be located convenient to the operator's work station(s).

3.2.5 Safety and health requirements. All parts, components, mechanisms, and assemblies furnished on the lathe whether or not specifically required herein, shall conform to all requirements of OSHA 29 CFR 1910 and ANSI 611.6. If a conflict arises between 29 CFR 1910 and ANSI B11.6, 29 CFR 1910 shall apply. Additional safety and health requirements shall be as specified (see 6.2). Covers, guards, or other safety devices normally furnished as standard on the manufacturer's commercial machine supplied to the commercial market shall be provided for the point of operation and all other parts of the lathe that present safety hazards. If point-of-operation guarding is influenced by workpiece size and configuration, the procuring activity shall provide part design data to the supplier of the machine (see 6.2). The supplier of the machine shall provide the special guarding which shall meet OSHA 29 CFR 1910.

3.2.6 Mecury restriction. The lathe shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

3.2.8 Environmental Protection. During the manufacture, operation, service, transportation, or storage of the machine and its components, materials hazardous to the ecological system as prohibited by applicable Federal, state, or local statutes in effect on the date of the contract shall not be used or emitted.

3.2.9 Lubrication. Means shall be provided to ensure adequate lubrication for all moving parts. Recirculating lubrication systems shall include a cleanable or replaceable filter. Means for determining oil level in each reservoir shall be by an oil level sight gauge, a low oil level warning light, or similar device and shall be in clear view of the operator's normal work station. An automatic lubrication system shall be provided for all ways and axis screws. Each lubricant reservoir shall have at least a 24-hour capacity and means for determining fluid level. All oil holes, grease fittings, and filler caps shall be readily accessible.

3.2.10 Interchangeability. All parts subject to replacement shall be manufactured to definite dimensions and tolerances. Replacement parts (except for hand-fitted parts such as ways and boxed bearings) shall be interchangeable without requiring modification. Replacement parts are those which are subject

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:02Z
Check the source to verify that this is the current version before use.

to wear or failure such as gears, bearings, shafts, feed screws, drive belts, pumps, and seals. Electrical replacement parts shall include, but not be limited to, motors, relays, fuses, switches, magnetics, pushbuttons, light bulbs, braking mechanisms, and solid-state devices. Tooling and accessories normally supplied as standard items shall also be considered replacement parts. Foreign-sourced replacement parts shall be interchangeable with domestically supplied components or be readily available from a domestic source.

3.3 Construction. The lathe shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion or failure.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Fastening devices. All fasteners shall be installed to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.3 Surfaces All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing,or soldering Welding, brazing, or soldering shall be employed only where specified in "the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.5 Painting. Unless otherwise specified (see 6.2), the paint color and application shall be in accordance with the manufacturer's commercial practice. All surfaces to be painted shall be cleaned of all foreign matter prior to paint application.

3.3.6 Threads. All threaded parts used on the lathe and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Gears. All gears shall conform to or exceed all provisions of AGMA 390.03a, AGMA 2000-A88, and AGMA 2001-B88 for the inch-pound system of measurement or ANSI/ISO 54 for the metric system of measurement. All gears shall be manufactured of a suitable ferrous material. The gears shall be of proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine. Load carrying gears in the drive train shall be hardened and ground steel. Working surface hardness shall be not less than Rockwell C-48.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:02Z
Check the source to verify that this is the current version before use.

3.3.8 Electromagnetic interference control. When specified (see 6.2), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystem(s) class and the emission and susceptibility requirements shall be as specified

3.4 Components. The lathe shall consist of not less than a bed and base, headstock, headstock spindle, spindle drive, spindle speed change mechanism, carriage, apron, cross slide, compound rest, and tailstock. In addition, all equipment necessary for the machine to meet the performance requirements stated herein shall be provided.

3.4.1 Bed and base. The lathe shall have a bed and base structure that supports the headstock, tail stock, carriage and other related equipment. The bed shall be a stress relieved ferrous casting with sufficient strength and rigidity to maintain accurate alignment of all the components in accordance with table III. The bed shall have V-ways or flat ways or a combination of “V” and flat ways to provide a sliding surface for the longitudinal movement of the carriage and to assure alignment of the headstock and tailstock. The bed ways shall be either cast integrally with the bed structure or replaceable way inserts, or a combination thereof. The bed ways for the carriage longitudinal movement shall be hardened to Rockwell C-50 if integral or Rockwell C-55 if inserted, and ground to minimize wear and maintain accuracy of alignment. Means shall be provided to lubricate the bed ways automatically or with a one-shot lubrication method. If a particular method of lubrication is required, it shall be as specified (see 6.2). The bed shall have a sufficient number of covered access openings to facilitate inspection, adjustment, and repair of components located within the structure. The base shall be provided with means for securing and leveling the machine on a foundation.

3.4.2 Chip and coolant pan. The chip and coolant pan shall allow coolant to drain from the chips and return to the coolant system for reuse. There shall be adequate clearance between the chip pan and the lathe bed to allow for chip removal.

3.4.3 Headstock. The lathe shall have a headstock that supports and aligns the spindle axis parallel with the bed ways. It shall contain all the gearing necessary to provide the required speeds; clutch and quick acting brake mechanism to start and stop the spindle rotation and a power takeoff to the gearbox for feeds and threading. Speed selections shall be by manually setting levers or dials on the headstock. The speed selections shall include a neutral position to allow free rotation of the spindle to facilitate speed changes and to allow use of a dial indicator in locating the workpiece. Spindle start and stop shall be by levers on the headstock and on the apron of the lathe. The headstock shall be constructed to support the spindle and gear shafts so as to prevent deflection that would affect accuracy under full operating and cutting loads. All load carrying surfaces of gears, clutch jaws and surfaces of sliding parts, shall have a surface hardness of not less than Rockwell C-48. When power to the spindle drive mechanism is transmitted through V-belts, means shall be provided to maintain proper belt tension. The headstock shall be oiltight and shall have automatic lubrication to all internal parts that require lubrication.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:02Z
Check the source to verify that this is the current version before use.

When specified (see 6.2), a temperature sensor shall be provided to monitor the temperature of the headstock lubricant and to shut the machine off at a preset temperature limit below the flashpoint of the oil vapor.

3.4.4 Headstock spindle. The lathe shall have a headstock spindle to provide accurate mounting and rotation of a chuck, faceplate, or collet. The spindle shall be supported by roller or ball type bearings that have a size, type and precision class suitable for the application. Means for adjusting thrust bearings shall be provided. The spindle nose shall conform to the requirements of ANSI B5.9 and unless otherwise specified (see 6.2) shall be the type and size specified in table I. The end of the hollow spindle shall have a standard taper as specified in table I and shall conform to the requirements of ANSI B5.10. The spindle nose shall be accurately machined to provide locating surfaces for lathe centers, faceplates and chucks. The inner and outer surfaces of the spindle nose on which work holding devices are mounted shall have a surface hardness not less than Rockwell C-40.

3.4.5 Spindle drive. The spindle shall be driven through either an all gear or combination gear and belt drive by an electric motor as specified in table I. The combination drive shall provide non-slip positive belt drive in the high speed range; the intermediate and low speeds shall be gear drive. The spindle shall operate in both directions of rotation.

3.4.6 Spindle speed change mechanism. The spindle speed mechanism shall be either the manual speed change typen or the variable speed type. When only one type spindle speed change mechanism is acceptable the mechanism shall be as specified (see 6.2). The shifting mechanism shall be designed so that speed change cannot occur unintentionally. Except for the variable speed drive, speeds shall be in geometric progression, plus or minus 10 percent through the range of speeds specified in table I. The manual speed change mechanism shall allow spindle speed changes by positioning gear shift levers or dials located on the front side of the headstock. Speeds shall be selected by positioning the levers as indicated on a conveniently located speed chart or by positioning a direct reading dial. The variable speed change mechanism shall permit spindle speed adjustments to compensate for varying machine conditions. Optimum speeds shall be obtained by gear range selection and by variable speed adjustment within the range. It shall be possible to adjust speed within a gear range while running. The actual spindle speed in revolutions per minute (rpm) shall be shown by a speed indicating device located convenient to the operator.

3.4.7 Feed thread gearbox. Unless otherwise specified (see 6.2), the lathe shall be equipped with a selective type, combination inch and metric gearbox to provide a full range of inch and metric feeds and threads as specified in table 1. The thread selections shall include the U.S. Standard Fine and Coarse Thread Series and Pipe Threads. The cross slide feed screw and the compound slide feed screw shall each be equipped with dials graduated to read measurements in both the inch system and the metric system. Power feeds and thread leads shall be provided by the totally enclosed quick change gearbox. Longitudinal and cross feeds shall be provided by a feed rod from the gearbox to a feed mechanism within the apron. Feeds shall be In either direction regardless of the direction of spindle rotation. Threading shall be done using a leadscrew from

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:02Z
Check the source to verify that this is the current version before use.

the gearbox and a split nut mechanism within the carriage apron. The lathe shall be able to cut either right-hand or left-hand threads regardless of the direction of spindle rotation. The gearbox shall be automatically lubricated. Safety clutches or other devices shall protect the feed/thread gearbox, feed rod, and leadscrew from accidental overload. An interlock shall prevent simultaneous engagement of the feed and thread mechanism. A feed and thread chart shall be mounted near the gear box as a ready reference for feed and thread selections. All feed and thread selections shall be shown without need for mathematical conversion. The end gearing shall have provision for use of transposing gears to obtain metric threads if an inch-type gearbox is required.

3.4.8 Carriage. The lathe carriage shall support the apron, cross slide, and compound rest. The carriage shall traverse longitudinally on the bed ways. Means shall be provided to maintain adjustment between sliding surfaces. A means shall be provided to securely lock the carriage in position on the bed when cross feeding. The carriage shall operate in both longitudinal directions under power, regardless of the direction of the spindle rotation. Way wipers shall be fitted to the carriage wings to prevent entrance of chips and abrasive materials between the way surfaces.

3.4.9 Apron. The lathe shall have an apron on the front side of the carriage which shall contain the feed mechanism, threading mechanism, operating controls, and lubrication system for the apron and carriage components. The operating controls shall include power longitudinal and cross feeds, automatically operated longitudinal feed stops, threading controls, and a spindle start and stop l Lever type operating controls such as the threading and feed engagement shall fully engage and disengage with smooth, positive action. The apron shall be double wall construction providing Inboard and outboard supports for both ends of gear shafts. Gearing in the apron provide hand and power longitudinal and cross feeds. An interlock shall be incorporated in the apron to prevent simultaneous engagement of the feed and half-nut. The feed rod and leadscrew shall be separate units.

3.4.9.1 Apron controlled leadscrew reverse. When specified (see 6.2), the lathe shall have an apron controlled leadscrew reversing mechanism to permit reversing the direction of the carriage without disengaging the leadscrew half-nut or changing the speed or direction of rotation of the spindle.

3.4.10 Cross slide. The cross slide shall be fitted to the carriage ways and shall have means to compensate for wear between the sliding surfaces. The cross slide shall be positioned by a precision feed screw which shall be protected from dirt and chips.

3.4.10.1 Compound rest. A compound rest shall be mounted on top of the cross slide. The base of the compound rest shall be graduated in one degree increments to not less than 90 degrees each side of a zero reference. The base shall swivel horizontally through 360 degrees. Means shall be provided to lock the compound at any angular setting. The compound slide feed screw shall be protected from dirt and chips.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:02Z
Check the source to verify that this is the current version before use.

3.4.11 Tailstock. The lathe shall have a tailstock for turning between centers and drilling operations. The tailstock shall be accurately fitted to the bedways and aligned with the headstock spindle. The tailstock shall be marked to indicate a zero position as aligned with the headstock spindle centerline and to indicate a set-over for turning shallow tapers without a taper attachment. The tailstock shall be fitted with wipers to prevent the entrance of dirt, coolant, and chips. The spindle internal taper shall be equipped for holding tang type tools having an American Standard Taper in accordance with ANSI B5.10 for self-holding tapers and shall eject tools when fully retracted. The spindle shall have a hardness of not less than Rockwell C-38 and shall be graduated in increments not greater than 1/16 inch or shall have a graduated micrometer feed dial for measuring spindle extension. The spindle shall be easily extended and retracted by handwheel to both limits of travel without jamming. The tailstock shall be easily moved along the bedways and have a quick acting clamp for locking the tailstock to the bed and a device for locking the spindle in any extended position without deflection. The tailstock mechanism and guide ways shall be provided with means for automatic or manual means of lubrication.

3.4.12 Coolant system. The coolant system shall be a complete unit for delivering a flow of coolant to the tool. The coolant system shall include a sump or reservoir, a power driven pump, and all necessary piping. The reservoir shall have a positive means to prevent chips from entering the pump. The system shall provide means for draining and cleaning. All materials used in the coolant system shall be compatible for use with all soluble and standard cutting oils. An adequate supply of coolant shall be continuously available at the cutting tool at all locations within the work envelope. An adjustable, flexile outlet tube shall provide final dispensing of coolant. A manually operated flow regulating valve shall be located near the tube outlet to allow adjustment of the stream of coolant between off and full flow, as required by the operator. A coolant off-on switch shall be located convenient to the operator.

3.4.13 Electrical system. The machine shall be wired to conform to NFPA 79. Unless otherwise specified (see 6.2), each machine shall be designed to operate on 230/460-volts, 3-phase, 60 Hertz (Hz). Unless otherwise specified (see 6.2), the lathe shall be initially wired for operation on 460 volts. Each machine shall have a fused safety disconnect switch or circuit breaker. An identified terminal for grounding the machine when it is installed shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in NFPA 79 for the disconnect fuse rating. The wiring in each machine shall have individual point-to-point color coded wires in conformance with NFPA 79. Diagrams/schematics shall clearly identify the individual, point-to-point color coded wire and wire termination point location in the electrical system.

3.4.14 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2), alternating-current (at) motors shall be designed to operate on 60-Hz. All motors shall conform to the requirements of NEMA MG1 and the motor enclosure shall meet the NEMA MG1 requirements for a drip proof enclosure. Each motor shall have an identification plate to identify

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:02Z
Check the source to verify that this is the current version before use.

manufacturer, model identification, serial number, voltage, amperage, horsepower, phase and frequency. Motor horsepower shall be the manufacturer's standard or standard optional horsepower as indicated on the manufacturer's machine design and construction drawing(s). It shall be unacceptable to substitute a motor larger than the manufacturer's standard to meet the requirements of table I.

3.4.15 Worklight. Unless otherwise specified (see 6.2), a worklight shall be provided. The worklight shall be a complete unit with all necessary equipment for mounting on the carriage, shall have a protective shield, an off-on switch and shall be adjustable to provide the maximum lighting of the work area. The lamp shall be shielded against dirt and chips and shall operate on no more than 130 volts. If a built-in lighting system is supplied the worklight described shall not be required.

3.4.16 Hour meter When specified (see 6.2), the lathe shall be equipped with an hour meter. "The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 99,999 hours in increments of not greater than 1 hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the 1athe.

3.5 Size and capacity. Unless otherwise specified (see 6.2), the size and capacity of the lathe shall meet the requirements of table I for the machine specified. Machine capacity shall be not less than the stated requirements of the size ordered. When a range is given, the required performance shall be from the stated minimum or less to the stated maximum or greater.

3.5.1 Physical size limitations. When specified (see 6.2), the physical size of the machine shall not exceed the height, width, and length restrictions specified by the procuring activity to ensure the machine will fit in its future operating location.

3.6 Alignment tolerances. Each 1athe shall meet the alignment and performance accuracies set forth in table III using the tests described in ANSI B 5.16, "Accuracies of Engine and Toolroom Lathes." Type I lathes shall meet or exceed the values found under the heading "Engine Lathes.' Type II lathes shall meet or exceed the values found under the heading "Toolroom Lathes."

3.7 Performance. The lathe shall be capable of meeting all the performance requirements specified herein when tested in accordance with 4.7.2 and 4.7.4.

3.7.1 Additional performance requirements Performance features required in addition to, or In lieu of, the requirements herein shall be as specified and fully described (see 6.2).

3.8 Standard equipment All standard equipment required to properly operate the lathe (such as special tools) shall be furnished. The equipment specified in 3.8.1 through 3.8.7 shall also be furnished as standard.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:02Z
Check the source to verify that this is the current version before use.

3.8.1 Lathe centers and center sleeve. One headstock center and one tailstock center shall be furnished. One headstock center sleeve shall be provided for adapting the taper in the spindle nose to the taper of the center supplied. The centers shall be hardened to Rockwell C-62 to C-66. The centers and the sleeve shall be ground to within 0.0003 inch total indicator reading for concentricity when inserted in the headstock and checked for concentricity.

3.8.2 Chuck and chuck jaws. The size and type of the chuck and the number of chuck jaws shall be as specified, (see 6.2). The chuck shall be a wrench operated, manual chuck, constructed of steel and shall mount on the spindle nose without the need for an adapter. The top jaws shall be the reversible, stepped type for internal and external gripping and shall be hardened. A self-ejecting chuck key shall be provided for each chuck. In all other aspects the chuck shall conform to the requirements of ANSI B5.8 for Class I, Medium Duty, Wrench Operated, Chucks.

3.8.3 Thread-chasing dial. A thread-chasing dial shall be furnished to allow disengaging of the half-nut and re-engaging during threading. The thread chasing dial shall be mounted on the carriage apron and be driven by the leadscrew.

3.8.4 Pastive thread stop. A positive thread stop shall be fitted to the crosslide to permit quick withdrawal and repositioning of the tool after each pass when thread chasing.

3.8.5 Wrenches. All special wrenches or tools that are required for operation or maintenance of the lathe and are normally furnished with the lathe shall be provided.

3.8.6 Tool post. The tool post shall be either a round single screw type with a rocker base, a square indexing turret type for holding at least four tool holders, or a quick change tool post capable of holding a tool holder in two positions perpendicular to each other on adjacent sides as specified, (see 6.2).

3.8.7 Micrometer carriage stop. A carriage stop shall mount on the front bed way and shall have a precision threaded micrometer screw graduated in thousandths of an inch for one inch movement.

3.9 Optional equipment. Optional equipment shall be furnished as specified and fully described (see 6.2). The optional equipment provided shall be functional without requiring modification of the equipment or machine.

3.10 Repair parts. Required repair parts shall be furnished as specified (see 6.2).

3.11 Marking on instruments, control panels, charts, and plates. All words on instruments, control panels, charts, and plates shall be in the English language. Characters shall be permanently marked In boldface on a contrasting background. All plates shall be corrosion resistant.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:02Z
Check the source to verify that this is the current version before use.

3.11.1 Lubrication plate. Unless otherwise specified (see 6.2), a lubrication plate shall be permanently and securely attached to each machine. Unless otherwise specified (see 6.2), the plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity Federal or military specification for each lubricant
Size and type of each lubricant filter (where applicable)

3.11.2 Nameplate. A nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2), the nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model number of the basic standard machine and a suffix identified in the manufacturer's permanent records.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Amp rating of largest motor
Short circuit/over-current protection rating
Contract number or order number
National Stock Number or Plant Equipment Code
Date of manufacture

3.12 Workmansh ip Workmanship of the lathe and its accessories shall meet all requirements specified herein

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance All items shall meet all requirements of sections 3 and 5. The inspection: set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:02Z
Check the source to verify that this is the current version before use.

does not authorize submissions of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in 4.7.1. .

4.4 First article inspection When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. The machine shall pass the examination and all tests to be accepted.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each lathe prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination in 4.6 the tests in 4.7.1 through 4.7.4.4 and the inspection in 4.8. The machine shall pass the examination, all tests, and the inspection to be accepted.

4.6 Examination. Each lathe shall be examined to determine compliance with all requirements of this specification.

4.7 Tests.

4.7.1 Test conditions. All tests shall be performed in an indoor facility with ambient conditions of 41 to 104 Fahrenheit and 20 to 95 percent relative humidity.

4.7.2 Operational test Each machine shall be operated at no load for not less than 1 hour. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.7.3 Alignment tolerance test Each machine shall be tested to determine compliance with the alignment tolerance requirements of 3.6. The alignment tolerance test shall be performed prior to and subsequent to the performance tests given in 4.7.4. Any machine adjustments required to comply with 3.6 during or after the performance test shall be cause for rejection.

4.7.4 Performance tests Each lathe shall be subjected to the following cutting tests to determine whether the performance under cutting loads is within the requirements of this specification. All performance tests shall be completed without making any adjustments to or performing any maintenance on the machine.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:02Z
Check the source to verify that this is the current version before use.

4.7.4.1 Round turning test. A test bar of 60,000 psi minimum tensile strength steel shall be held in a chuck and turned for a length of approximately two inches at the rpm and feed rate based on the recommended surface cutting speed for the material and tool being used. The finish cut shall be conducted in accordance with test number 21 of ANSI B5.16 and shall meet the tolerances as stated in table III of this specification under the proper category for the type lathe specified. In addition, the surface finish shall be no more than 32 microinches,

4.7.4.2 Cylindrical turning test. A test bar of 60,000 psi minimum tensile strength steel shall be rough turned to the configuration of the specimen in test number 22 of ANSI B5.16 for cylindrical turning. A finish cut using carbide tooling shall be made over the two collars in one pass without adjustment between collars. The turned collars shall be within the accuracy tolerance specified in test number 22 of table III under the proper category for the type lathe ordered. The test part shall then be prepared for turning between centers as described in test number 23 of ANSI B5.16. A cut shall be made over the two collars and the diameters shall be within the tolerance for turning cylindrical between centers in test number 23 of table II. spindle speeds and feeds shall be based on the recommended surface cutting speed for the material and carbide tooling being used.

4.7.4.3 Roughing test. A square specimen of 60,000 psi minimum tensile strength steel with 3-inch sides, and 16 inches in length shall be rough-turned between centers by two or more cuts; depth of cut 0.250 inch, and a cutting speed of approximately 40 RPM, and feed rate of approximately 0.035 inches per revolution. The maximum diameter of the bar at the tailstock and the maximum diameter of the bar measured 10 and 15 inches from the tailstock end of the bar, shall not vary by more than plus or minus 0.001 inch. The kickback of the carriage relative to the bed shall be measured and shall not exceed 0.001 inch during any single cut. The deflection of the tailstock spindle shall not exceed 0.002 inch, measured 1/8-inch from the nose of the spindle during the roughing test.

4.7.4.4 Additional performance tests. Performance tests required in addition to or in lieu of the requirements given shall be as specified and fully described (see 6.2).

4.7.5 Electromagnetic interferenc control test Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.8 using the procedures given in MIL-STD-462.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-18058.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:02Z
Check the source to verify that this is the current version before use.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but Is not mandatory.)

6.1 Intended use. The engine lathes and tool room lathes covered by this specification are intended for use in any machine shop for production operations or tool room type operations where precision tolerances are required in producing work of a diverse nature. The lathes are suitable for heavy duty roughing and light finishing operations such as turning, boring, facing, drilling, and threading using high speed steel and carbide tooling.

6.2 Acquistion requirements l Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, class, and size required (see 1.2).

c* Issue of DODISS to be cited In the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

d. First article, if required (see 3.1).

e. System of units for measuring and indicating devices (see 3.2.1).

f. Digital readout system, if not required (see 3.2.1.2).

9* Specify any special digital readout system features, if required (see 3.2.1.2).

h. Additional safety and health requirements, if required (see 3.2.5).

j. Specify configuration of additional point-of-operation guarding, if required (see 3.2.5).

j. Painting, if different (see 3.3.5).

k. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.8).

l. Specify type of bed way lubrication, automatic or one-shot (see 3.4.1).

m. Headstock temperature sensor, if required (see 3.4.3).

n. Spindle nose, if different (see 3.4.4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:02Z
Check the source to verify that this is the current version before use.

o* Spindle speed change mechanism type if only one type is acceptable, specify which (see 3.4.6).

p. If either English or metric gearbox is required, specify which (see 3.4.7).

q. Apron controlled leadscrew reverse, if required (see 3.4.9.1).

r. Electrical requirements, if different (see 3.4.13).

s. Initial voltage, if different (see 3.4.13).

t. Motors, if different (see 3.4.14).

u. Worklight, if not required (see 3.4.15).

v. Hour meter, if required (see 3.4.16).

w. Size and capacity, if different (see 3.5).

x. Specify physical size limitation, if required (see 3.5.1).

y. Additional performance features, if required (see 3.7.1).

z. Specify chuck and chuck jaws required, (see 3.8.2).

aa. Specify type of tool post required, (see 3.8.6).

bb. Specify and fully describe optional equipment, If required (see 3.9).

cc l Specify applicable repair parts, if required (see 3.10).

dd. Lubrication plate, if not required, (see 3.11.1).

ee. Information to be provided on lubrication plate, if different (see 3.11.1).

ff. Nameplate, if different (see 3.11.2).

gg. Inspection conditions, if different (see 4.3).

hh. First article inspection, if different (see 4.4).

ii. Quality conformance inspection, if different (see 4.5).

jj. Performance tests, if different; fully describe (see 4.7.4.4).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The

Source: https://assist.dla.mil -- Downloaded: 202610-19T16:02Z
Check the source to verify that this is the current version before use.

applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that on1y essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DO Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
.	.	.	.
.	.	.	.

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.3.1 Technical data. It should be specified in the “Remarks” section of the DD Form 1423 that all technical data ordered shall be written in the English language.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor’s current inventory and the number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Definitions.

6.5.1 Inch-pound units. Inch-pound units are a system of measures based on the yard and pound commonly used in the United States of America and defined by the National Institute of Standards and Technology. Inch-pound units having the same names in other countries may differ in magnitude.

6.5.2 Metric units. Metric units are a system of basic measures defined by the International System of Units based on "Le Systeme International D’Unites (SI)," of the International Bureau of Heights and Measures. These units are described in ASTM E 380 and IEEE 268 (MIL-STD-961).

6.6 Measurement system. In this specification, all measurements, dimensions, sizes, and capacities are given in inch-pound units. These measurements may be converted to metric units through the use of the conversion factors and methods specified in FED-STD-376.

Source: https://assist.dla.mil -- Downloaded: 2026l0-19T16:02Z
Check the source to verify that this is the currentversion before use.

6.7 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the-changes.

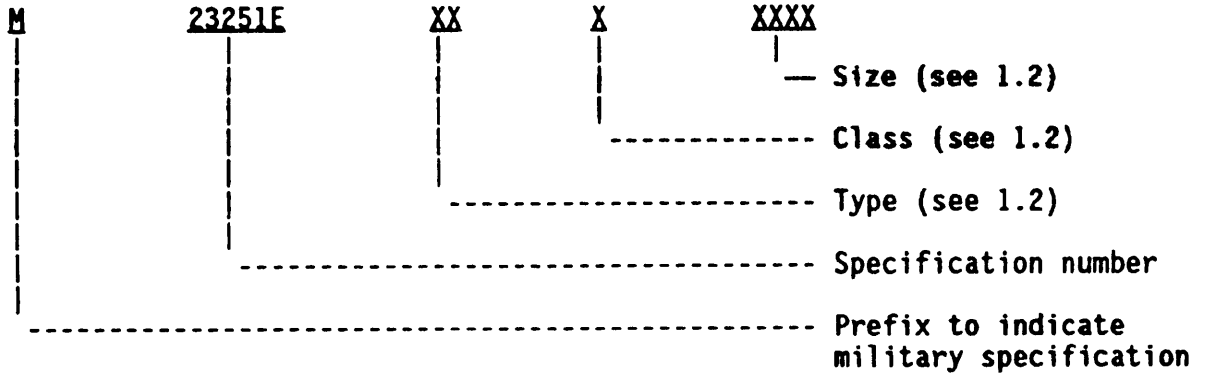
6.8 Safety and health requirements. OSHA 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants, and heat) of the environment in which a machine will operate. It does not limit the hazard level of individual machines in an operating environment. The procuring activity should analyze the existing hazard level in the proposed operating environment and specify additional requirements necessary to integrate this new machine into its future environment.

6.9 Training. Training required by the procuring activity should be provided as specified in the contract.

6.10 Warranty. Warranty requirements should be as specified by the procuring activity in the contract.

6.11 Inspection location. The contractor should identify in his response to the solicitation the location where inspection and tests are to be performed.

6.12 Part or Identifying Number (PIN). The PIN to be used for machines acquired to this specification are created as follows:



(NOTE: If Roman numerals (I, II, etc.) are used for class or type, add a note stating that in the PIN, Arabic numbers shall be used.)

6.13 Subject term (keyword) listing.

Turning
Manual
Military specification
Variable speed

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:02Z
Check the source to verify that this is the current version before use.

TABLE I. Sizes and capacities.

Capacities Dimensions in inches (except as noted)	Type I and Type II Size 2516 Class 1	Type I Size 2516 Class 2	Type I and Type II Size 3220	
Capacities:				
Swing over bed and carriage wings	25	27	33	36
Swing over cross slide minimum	16	18	27	24
Center distance, basic	48	54	48	48
**Optional center distance increments	24	24	24	24
Drive motor horsepower, manufacturer's				
Standard for the speed range specified	25 or 30	15 or 20	30 or 40	40 or 50
Headstock:				
Spindle type and size	6"-D1 or 11"-A2	8" D1	15"A1 or 11"D1	11"D1 or 15 A1
Diameter of thru hole (minimum)	2	3	2-3/16	3-1/2
Center size, morse taper	#4 #5 or #6	#5	#6	#6
number of speeds or infinitely variable	12	18	18	36
Speed range, RPM	30 to 650	30-1580	10 to 850	8 to 1000
Optional speed range	40 to 1000 or 40 to 1500			
Carriage:				
Cross slide travel (minimum)	13	12	20	20
Top slide travel (minimum)	5	3-1/2	7	6
Inch Gearbox:				
Number of feeds and threads (minimum)	48	60	55	48
Range of feeds	0.006-0.080	0.0021 -0.1242	0.005 to 0.070	0.003 to 0.160
Range of threads	2 to 64	2-3/4 -160	1 to 60	1 to 64
Metric Gearbox:				
Feed and thread selections, number of (minimum)	48	60	48	48
Feed range, mm/rev.	0.056 to 2.011	0.056 to 2.011	0.065 to 4.13	0.8 to 4.3
Thread range, lead in mm	0.250 to 16	0.250 to 16	0.5 to 32	1 to 56
Combination Inch and Meteric Gearbox:				
Feed and thread selections, number of (minimum)	48	48	48	48
Feed range, IPR	0.006 to 0.080	0.006 to 0.060	0.0005 to 0.160	0.003 to 0.160
mm/rev	0.090 to 2.011	0.090 to 2.011	0.065 to 4.13	.08 to 4.3
Thread range, per inch	2 to 64	2 to 64	1 to 64	1/2 to 28
lead in mm	0.250 to 16	0.250 to 16	0.065 to 4.13	1 to 56
Tailstock:				
Spindle diameter	4	3	6	6-1/2
Spindle travel (minimum)	7	6	11	11
Center size, morse taper	#5 or #6	#5	#6	#6
Faceplate diameter, large	18	21	21	
Chuck size 3-jaw	16	16	16	
Chuck size 4-jaw	18	21	18	

NOTE: ** Optional increments are in addition to basic center distance of the type, size, and class machine specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:02Z
Check the source to verify that this is the current version before use.

TABLE II. Sizes and capacities of accessory equipment.

(Dimensions in inches)

Characteristics	Size 2516		Size 3220
Optional equipment:	Class 1	Class 2	Class 1 and 2
Taper attachment:			
Taper per foot and lenght turned:			
Plain bearing type	0 to 3-1/2 by 18	0 to 3-1/2 by 21	0 to 3-1/2 by 21
Optional, anti-friction bearing type	0 to 4 by 18	0 to 6-1/2 by 24	0 to 6-1/2 by 24
Steady rest capacity:			
Standard rest capacity	1 to 6	1 to 7	2 to 9
Optional oversize rest capacity from maximum standard capacity:	12	14	19
Follow rest capacity	1 to 4-1/2	1 to 6	1 to 6

NOTE: When a range is shown, the required performance is from the stated minimum or less to the stated maximum or greater.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:02Z
Check the source to verify that this is the current version before use.

TABLE III. Alignment and performance accuracies.

ANSI B5.16 test number	Test (Dimensions in inches)		Toolroom lathe	Engine lathe
1	Bed level, transverse direction, in 12 inches		±0.00024	±0.00024
2	Bed level, longitudinal direction, in 12 inches		within 0.00024	within 0.00024
3	Tailstock way alignment, in 12 inches		within 0.00024	within 0.00024
4	Spindle center runout		0 to 0.0004 TIR	0 to 0.0004 TIR
5	Spindle nose runout		0 to 0.0003 TIR	0 to 0.0004 TIR
6	Cam action of spindle		0 to 0.0003 TIR	0 to 0.00075 TIR
7	Spindle taper runout, at end of 12-inch test bar		0 to 0.0006 TIR	0 to 0.0012 TIR
		at end of spindle nose	0 to 0.0003 TIR	0 to 0.0004 TIR
8	Headstock alignment vertical, in 12 inches		0 to 0.0005	0 to 0.0008
9	Headstock alignment horizonal, in 12 inches		0 to 0.0002	0 to 0.0008
10	Tailstock spindle alignment horizontal in 4 inches		0 to 0.0004	0 to 0.0004
11	Tailstock spindle alignment vertical, in 4 inches		0 to 0.0008	0 to 0.0008
12	Tailstock taper alignment horizontal, in 12 inches		0 to 0.0004	0 to 0.0008
13	Tailstock taper alignment vertical, in 12 inches		0 to 0.0004	0 to 0.0012
14	Vertical alignment of head and tail centers, high at tailstock		0 to 0.008	0 to 0.0012
15	Readings for base length bed taken with lead screw stationary:	Vertical	0 to 0.004	0 to 0.006
		Horizontal	0 to 0.004	0 to 0.006
	Alignment of half-nut horizontal or vertical		0 to 0.006	0 to 0.006
16	Lead screw cam action		0 to 0.0002	0 to 0.0004
17	Cross-slide alignment, on 12-inch diameter concave only		0 to 0.0005	0 to 0.001
18	Face plate runout	on diameter	0 to 0.0005	0 to 0.001
		on face at diameter	0 to 0.001	0 to 0.0015
19	Chuck runout	face and periphery	0 to 0.003	0 to 0.003
		face of steps	0 to 0.003	0 to 0.003
	Bar test 3 inches from end of jaw		0.003 maximum	0.003 maximum
20	Collet chuck runout 1 inch from collet chuck		0 to 0.001	0 to 0.001
21	Lathe must turn round with work mounted in chuck		0 to 0.0002	0 to 0.001
22	Lathe must turn cyclindrical, 12-inch workpiece mounted in a chuck		0 to 0.0008	0 to 0.0012
23	Lathe must turn cylindrical, 12-inch workpiece mounted on centers		0.0004	0.0008
24	Leadscrew, lead per 12 inches		+0.001	+0.002
	lead in any 4 inches		+0.0004	+0.0007
25	Backlash on cross-feeds screw		0.004	0.005
	on compound rest screw		0.004	0.005

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:02Z
Check the source to verify that this is the current version before use.

Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Army - AR
Navy - YD
Air Force - 84

User activities:
Navy - MC

Preparing activity:
DLA - IP

Project (3416-0223)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:02Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEFENSE LOGISTICS AGENCY
Headquarters
Defense Industrial Plant Equipment Center
Memphis, Tennessee 38114-5051

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO 4966 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEFENSE LOGISTICS AGENCY

Commander
Defense Industrial Plant Equipment Center
ATTN: DIPEC-SSM
Memphis, Tennessee 38114-5051

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:02Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-L-23251E	Lathes, Engine and Toolroom, 2516 and 3220

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6 REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:02Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:02Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:02Z
Check the source to verify that this is the current version before use.